Biological Counter-Curvature: An Information–Cosserat Framework for Vertebral Geometry and Adolescent Scoliosis

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Abstract

Why do vertebrate spines maintain S-curves against gravity? Existing models prescribe geometry but do not derive it from first principles. We present the Information-Elasticity Coupling (IEC) framework, where a developmental information field $I(s)$ modifies Cosserat rod rest curvature and stiffness, selecting the S-curve as the energetic minimum. Phase diagram analysis reveals three regimes; the cooperative regime matches normal human anatomy (Cobb angles 20–50°). AlphaFold structural analysis of 23 mechanically relevant proteins reveals a 34% structural cost asymmetry between error-correction and energy-supply subsystems. This creates a Demand–Supply Anisotropy Gap, predicting an Energy Deficit Window during adolescent growth driven by six distinct molecular mechanisms of energy supply failure beyond hunger. A predicted VIM cascade explains irreversible progression. This framework resolves five long-standing clinical questions regarding adolescent idiopathic scoliosis etiology and generates 12 quantitative, falsifiable predictions for future experimental and clinical validation.

Model Summary: The Information–Elasticity Coupling (IEC) Framework

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The Clinical Problem: Why does the spine buckle laterally (scoliosis) during the adolescent growth spurt?

The Core Hypothesis: The S-shaped human spine is not a passive mechanical structure but an active *counter-curvature* maintained by metabolic energy and guided by developmental information fields (e.g., HOX genes).

The Mechanism:

- **Information Field $I(s)$**: Defines the target healthy curvature at each spinal level s.
- **Metabolic Cost (L^4 vs L^2)**: Maintaining this target shape against gravity requires energy that scales with the spine's length to the fourth power (L^4), while nutrient supply scales only with surface area (L^2).
- **Energy Deficit Window**: During rapid adolescent growth, the system hits a metabolic deficit, forcing a down-regulation of expensive mechanosensors (e.g., PIEZO2).
- **Vector Mismatch & Buckling**: Loss of sensory fidelity creates a ”vector mismatch,” drastically reducing the spine's lateral stiffness. The system minimizes energy by collapsing into a scoliotic spiral.

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1 Introduction

Life on Earth has evolved within the unyielding context of a constant gravitational field. Gravity acts not merely as a mechanical load but as a ubiquitous morphogenetic cue. A 70 kg adult exerts approximately 686 N of axial force through the spine in upright posture. Maintaining this non-geodesic, load-bearing geometry requires continuous metabolic expenditure—an estimated 100 W just for upright standing. This presents a classical biological question: how does genetic information specify and maintain a three-dimensional geometry that actively resists physical forces?

Current theories explain aspects of this process but face critical limitations. Morphoelastic rod theory models how growth tensors prescribe rest curvature but does not directly link to developmental genetics or metabolic energy balance [1]. Biomechanical scoliosis models, such as the Hueter-Volkmann effect, describe a vicious cycle of asymmetric loading and growth modulation *after* a deformity initiates but cannot explain the etiologic origin [2]. Hormonal theories based on estrogen or melatonin offer epidemiological correlations but lack a unifying mechanical mechanism. Proprioceptive control theories, such as those implicating PIEZO2, explain alignment maintenance but do not provide a quantitative framework for adolescent vulnerability [3]. A unifying quantitative framework connecting genetic information, tissue mechanics, metabolic energy, and clinical deformity does not yet exist.

To resolve this, we present the Information-Elasticity Coupling (IEC) framework. We show that coupling a developmental information field $I(s)$ to Cosserat rod rest curvature and stiffness selects the characteristic spinal S-curve as an energy minimum against gravity. While our framework connects developmental patterning to macroscopic geometry, the mechanistic basis lies in protein-level biophysics. HOX genes regulate the synthesis of mechanosensory proteins (e.g., integrins) and structural proteins (e.g., collagens, proteoglycans), which compete for limited metabolic resources during growth. This competition creates thermodynamic instability windows where the system becomes sensitive to perturbations. Furthermore, information-elasticity coupling is fundamentally tensorial: tissue response depends on the alignment between developmental gradient directions and mechanical stress vectors. These protein-level and geometric details provide the mechanistic foundation for IEC.

This approach reveals that the S-shape is maintained at a thermodynamic cost. AlphaFold structural analysis of 23 proteins reveals a 34% structural cost asymmetry (Demand-Supply Anisotropy Gap) between error-correction and energy-supply subsystems. This creates a predictable Energy Deficit Window during adolescent growth, characterized by six distinct molecular mechanisms of energy supply failure beyond caloric insufficiency. The framework explains the irreversible progression of adolescent idiopathic scoliosis (AIS) via a VIM collapse cascade and generates 12 quantitative, testable predictions.

AIS affects approximately 3% of adolescents worldwide. Current treatments rely heavily on bracing and invasive fusion surgery because the fundamental etiology remains unknown. The IEC framework provides the first quantitative etiological theory of AIS, introducing patient-stratification metrics (such as the χ_κ parameter) with the potential to predict scoliotic progression. While this work relies on theoretical and computational modeling rather than direct wet-lab experimentation, the mathematical framework and emergent predictions offer a new paradigm for understanding and clinically managing spinal geometry.

2 Theory

We propose that the robust S-shaped geometry of the spine arises not from passive mechanical equilibrium under gravity, but from an active *counter-curvature* mechanism driven by developmental information. We formalize this using an Information–Elasticity Coupling (IEC) framework, where a scalar information field $I(s)$ modifies the effective geometry and energetics of a

Cosserat rod.

2.1 The Counter-Curvature Hypothesis

The central problem of vertebrate morphogenesis is the translation of a one-dimensional genetic code into a robust three-dimensional geometry that can withstand environmental loads. In the case of the human spine, this challenge is acute: the column must support the head and trunk against the relentless downward pull of gravity ($g \approx 9.81$ m/s^2) while maintaining flexibility for locomotion. Classical structural mechanics predicts that a flexible column clamped at the base and subjected to its own weight will buckle or sag into a monotonic C-shape [1]. Yet, the healthy human spine exhibits a complex, alternating S-shape (lordosis and kyphosis).

This anomaly suggests that the S-shape is not a passive equilibrium but an active *Counter-Curvature*—a geometric standing wave maintained by the continuous expenditure of metabolic energy against the gravitational field. This hypothesis bridges two foundational concepts in biology: the *Morphogenetic Field* of developmental biology [4, 5] and the *Mechanostat* of tissue physiology [6].

The Information Field as a Target Metric. Developmental patterning establishes a "Positional Information" field along the anterior-posterior axis, encoded by the nested expression of HOX genes [7]. We posit that this information field $I(s)$ does not merely specify cell identity but encodes a *target metric*—a preferred intrinsic curvature $\kappa_{rest}(s)$ that the tissue strives to achieve. This is analogous to the "growth field" concepts in plant morphogenesis [8] but applied to a load-bearing animal structure. The spine is thus a "smart beam" that continuously computes the error between its actual geometry (sensed via proprioception) and its genetic target.

Active Maintenance and the Hydrostatic Skeleton. Achieving this target shape against gravity requires work. The spine functions partly as a *hydrostatic skeleton* [9], where the nucleus pulposus of the intervertebral disc acts as a pressurized hydraulic element. The maintenance of this pressure—and thus the maintenance of the S-shape—depends on the balance between osmotic swelling pressure (driven by proteoglycan synthesis) and the mechanical stress of gravity. This active maintenance constitutes a biological implementation of *optimal feedback control* [10], where the system minimizes a cost function comprising both "geometric error" (deviation from the HOX-specified shape) and "metabolic effort" (ATP consumption).

2.2 Rod Geometry and Kinematic Parameterization

We model the human spine as a Cosserat rod, defined by a centerline curve $\mathbf{r}(s,t) \in \mathbb{R}^3$ parameterized by arc length $s \in [0, L]$. This dimensionality reduction from 3D elasticity to a 1D director theory is justified by the high aspect ratio of the vertebral column ($L/d \approx 13$). To each point s, we attach an orthonormal material frame $\{\mathbf{d}_1(s,t), \mathbf{d}_2(s,t), \mathbf{d}_3(s,t)\}$ describing the cross-sectional orientation [1, 11]. The director $\mathbf{d}_3(s,t)$ aligns with the tangent to the centerline in the absence of shear, while $\mathbf{d}_1(s,t)$ and $\mathbf{d}_2(s,t)$ define the principal axes of the vertebral cross-section. The kinematics are governed by the evolution equations:

$$\mathbf{r}'(s) = \mathbf{v}(s) = \mathbf{d}_3(s) + \boldsymbol{\varepsilon}(s), \tag{1}$$

$$\mathbf{d}_i'(s) = \mathbf{u}(s) \times \mathbf{d}_i(s), \tag{2}$$

where $\mathbf{v}(s)$ is the tangent vector, $\boldsymbol{\varepsilon}(s)$ represents shear and extension strains, and $\mathbf{u}(s) = \kappa_1\mathbf{d}_1 + \kappa_2\mathbf{d}_2 + \tau\mathbf{d}_3$ is the Darboux vector encoding the flexural curvature components $\kappa_{1,2}$ and torsional twist τ. Physically, we identify $s = 0$ with the sacrum (clamped boundary condition: $\mathbf{r}(0) = \mathbf{0}, \mathbf{d}_i(0) = \mathbf{e}_i$) and $s = L$ with the cranium (free boundary condition), representing the aggregate mechanical axis of the vertebral column.

2.3 Developmental Information Fields from Morphogenetic Patterning

We define a scalar information field $I(s,t) \in [0,1]$ representing the coarse-grained expression level of anterior-posterior patterning morphogens (e.g., HOX genes). This field serves as a coarse-grained representation of the complex gene regulatory networks governing axial patterning. Following the bimodal Gaussian parameterization (Methods Eq. 19), we model $I(s)$ with peaks at the cervical ($A_c = 0.5, s_c = 0.80$) and lumbar ($A_l = 0.7, s_l = 0.25$) enlargements, corresponding to regions of high vertebral identity specification. The spatial gradient ∇I acts as a "curvature generator" via the geometric coupling χ_κ:

$$\boldsymbol{\kappa}_{\text{rest}}(s) = \boldsymbol{\kappa}_{\text{gen}} + \chi_\kappa \nabla I(s). \tag{3}$$

We quantify the information content of this patterning field using the local Shannon entropy $\mathcal{H}[I]$, which bounds the precision of the target metric specification:

$$\mathcal{H}[I] = -\int I(s) \log I(s)\, ds. \tag{4}$$

Simultaneously, the field exerts a stiffness via the morphomechanical coupling χ_M, generating the active biological moment $\mathbf{M}_{bio}$ to oppose gravity:

$$\mathbf{M}_{bio} = \chi_M(\boldsymbol{\Lambda} \cdot \nabla I). \tag{5}$$

Stability is quantified by the Bio-Gravitational Number $\mathcal{B}_g$:

$$\mathcal{B}_g = \frac{\chi_M \langle |\nabla I| \rangle}{\rho A g L^2}. \tag{6}$$

When $\mathcal{B}_g > 1$, the organism dominates gravity. Normal human anatomy clusters around $\mathcal{B}_g \approx 1.2 \pm 0.3$. This framework is physically grounded in the structural specificity of HOX proteins.

2.4 Information-Elasticity Coupling and Effective Energy

The central hypothesis of the IEC framework is that developmental information modifies the *effective* energy landscape of tissue deformation. We define an *effective metric* $d\ell^2_{\text{eff}}$ that encodes target geometry via a local dilation factor $g_{\text{eff}}(s)$. We use "effective metric" strictly as a computational metaphor for how information modifies the energy landscape, without invoking literal modification of physical spacetime.

$$d\ell^2_{\text{eff}} = g_{\text{eff}}(s)\, ds^2 = \exp\left[2\left(\beta_1 \tilde{I}(s) + \beta_2 \frac{\partial \tilde{I}}{\partial s}\right)\right] ds^2, \tag{7}$$

where $\tilde{I}$ is the normalized information field and $\beta_{1,2}$ are dimensionless coupling constants.

The mechanics of the biological rod are governed by the minimization of a total potential energy $\mathcal{E}_{\text{total}}$ that couples bending, torsion, and extension to the information field and gravity. For a Cosserat rod with centerline $\mathbf{r}(s)$, the full energy functional is:

$$\mathcal{E}_{\text{total}} = \int_0^L \left[\frac{1}{2} B_{\text{eff}}(\kappa - \kappa_{\text{rest}})^2 + \frac{1}{2} C_{\text{eff}} \tau^2 + \frac{1}{2} K_{\text{eff}} \varepsilon^2 + \rho A \mathbf{r} \cdot \mathbf{g}\right] ds, \tag{8}$$

where:

- $B_{\text{eff}}(s) = E_0 I_{\text{area}}(1 + \chi_E I(s))$ is the information-modulated bending stiffness.
- $C_{\text{eff}}(s) = G_0 J(1 + \chi_C I(s))$ is the torsional stiffness (G_0: shear modulus).
- $K_{\text{eff}}(s) = E_0 A(1 + \chi_A I(s))$ is the extensional stiffness.

- $\kappa_{\text{rest}}(s) = \kappa_0 + \chi_\kappa \partial_s I$ is the information-driven target curvature.
- τ and ε are the local torsion and strain, respectively.

The weighting $w(I) = (1+\chi_E I(s))$ penalizes deviations from the information-prescribed counter-curvature more heavily in high-identity regions (e.g., cervical/lumbar junctions).

2.5 Cosserat force and moment balance

The equilibrium configuration is found by finding the stationary state of the total potential energy. For a static rod subject to gravity $\mathbf{f}_g = \rho A \mathbf{g}$ and active moments induced by the information field gradients, the balance equations for internal force $\mathbf{n}$ and moment $\mathbf{m}$ are:

$$\begin{aligned} \mathbf{n}'(s) + \mathbf{f}_g &= \mathbf{0}, \\ \mathbf{m}'(s) + \mathbf{r}'(s) \times \mathbf{n}(s) + \mathbf{m}'_{\text{info}}(s) &= \mathbf{0}, \end{aligned} \tag{9}$$

where $\mathbf{m}_{\text{info}}$ represents the active couple.

Boundary Conditions: To model the human spine, we impose clamped-free boundary conditions. The sacral base ($s = 0$) is rigidly fixed, while the cranial tip ($s = L$) is free of external loads:

$$\begin{aligned} \mathbf{r}(0) = \mathbf{0}, \quad \mathbf{d}_i(0) &= \mathbf{e}_i \quad \text{(Clamped Base)} \\ \mathbf{n}(L) = \mathbf{0}, \quad \mathbf{m}(L) &= \mathbf{0} \quad \text{(Free Tip)} \end{aligned} \tag{10}$$

where $\{\mathbf{e}_i\}$ is the laboratory frame.

2.6 Mode selection and spinal geometry

The transition from a sagging C-shape to a counter-curved S-shape can be analyzed as a spectral shift in the rod's eigenmodes. In the linearized limit, small transverse deflections $y(s)$ satisfy:

$$\mathcal{L}_{\text{IEC}}[y(s)] = \frac{d^2}{ds^2}\left(B_{\text{eff}}(s)\frac{d^2 y}{ds^2}\right) - \frac{d}{ds}\left(N(s)\frac{dy}{ds}\right) = \lambda_n y_n(s), \tag{11}$$

where $N(s) = \int_s^L \rho A g \, ds'$ is the axial tension due to gravity. The information field modifies the stiffness landscape $B_{\text{eff}}(s)$. For a uniform beam ($\chi_E = 0$), the lowest eigenvalue λ_0 corresponds to a monotonic C-shaped mode. As χ_κ and χ_E increase, the effective stiffness at the peaks of $I(s)$ (cervical and lumbar regions) increases, making the C-mode energetically unfavorable. At a critical coupling strength χ_{crit}, a *mode crossing* occurs where the S-shaped mode (characterized by a second zero-crossing in curvature) becomes the ground state (λ_0). This selection mechanism operates independently of specific initial conditions, explaining the universal tendency toward S-shaped postures in upright vertebrates.

The energy functional $\mathcal{E}_{\text{total}}$ (Eq. 8) describes the potential energy landscape of the spine at equilibrium. However, the biological spine is not a conservative system: it is a *dissipative structure* [12] that requires continuous metabolic expenditure to maintain its S-shaped counter-curvature against gravitational loading.

We formalize this by defining a *metabolic power density* $\dot{\mathcal{F}}$ that quantifies the rate of free energy dissipation required to sustain the counter-curvature:

$$\dot{\mathcal{F}} = \int_0^L \left[\eta_p \left|\frac{\partial \kappa}{\partial t}\right|^2 + \eta_a \left(\kappa(s) - \kappa_{\text{passive}}(s)\right)^2 + \Gamma_m(s)\right] ds, \tag{12}$$

where:

- η_p is the proprioceptive feedback dissipation coefficient (W/m), representing sensory error correction.
- η_a is the active moment maintenance coefficient (W/m), capturing paraspinal muscle contraction required to hold the spine away from its passive C-shape.
- $\Gamma_m(s)$ is the basal tissue maintenance cost density (W/m).

The second term represents the standing metabolic cost of countercurvature. For a passive beam, this term vanishes; the S-curve imposes a non-zero thermodynamic tax.

Scaling with spinal length. The gravitational moment acting on the spine scales as $M_g \sim \rho A g L^2$. The active biological moment must match this, so the metabolic power required to sustain the S-shape scales as:

$$P_{\text{counter}} \sim \eta_a \rho A g L^2 \cdot \langle |\kappa_{\text{IEC}} - \kappa_{\text{passive}}|^2 \rangle, \tag{13}$$

During the adolescent growth spurt, L increases from approximately 0.35 m to 0.45 m, implying a 1.65-fold to 2.13-fold increase in required metabolic power. To formalize this transition, we introduce the **Thermodynamic Instability Ratio** $R(t)$:

$$R(t) = \frac{\text{Demand}(L^4 \langle \kappa^2 \rangle)}{\text{Supply}(L^2)} = \xi L^2, \tag{14}$$

where ξ is a coupling coefficient reflecting the avascular diffusion efficiency of the intervertebral disc. We posit that the Countercurvature mechanism undergoes a **Thermodynamic Bifurcation** when $R(t) > R(t)_{crit}$, creating a transient *energy deficit window.*

2.7 Protein-Level Biophysical Foundations and Thermodynamic Constraints

The Information-Elasticity Coupling (IEC) framework links developmental patterning to organ-scale geometry, but that link requires an explicit mechanistic bridge from gene regulation to tissue mechanics. We formalize IEC as a five-level cascade: (i) protein sequence and structure, (ii) single-molecule mechanics, (iii) fibril assembly, (iv) tissue-scale homogenized elasticity, and (v) whole-spine Cosserat equilibrium.

Mechanosensory vs. Growth Protein Dichotomy. A central mechanistic distinction exists between mechanosensory proteins and growth/structural proteins. Mechanosensory proteins (e.g., PIEZO2, Lamin A/C, integrins) act as "sensors" and "actuators" of the countercurvature system. They possess high structural anisotropy, extended conformations, and require high continuous ATP expenditure for trafficking and phosphorylation cycling. Their effective response time is fast (seconds to minutes). Growth/structural proteins (e.g., collagens, proteoglycans, SIRT1) constitute the "supply" side, defining the baseline material architecture. They are generally more compact, have amortized energetic costs after deposition, and operate over slower timescales (hours to days).

Thermodynamic Instability Windows. These classes are dynamically coupled through a shared metabolic pool. We represent the energy budget per unit tissue volume as:

$$E_{\text{total}}(t) = E_{\text{mechano}}(t) + E_{\text{growth}}(t) + E_{\text{metabolic}}(t), \tag{15}$$

During adolescence, E_{growth} rises with growth velocity, while effective supply can remain transport-limited. The result is a transient deficit in control-relevant energy allocation. We quantify this imbalance using an instability ratio based on growth versus adaptation velocities:

$$\Lambda(t) = \frac{v_{\text{growth}}(t)}{v_{\text{adapt}}(t)}, \tag{16}$$

When $\Lambda(t) > \Lambda_{\text{crit}}$, the system enters an unstable regime where $\partial\kappa/\partial\varepsilon \gg 1$. In this Energy Deficit Window, perturbations that would ordinarily be corrected are amplified because corrective sensing lags behind growth-driven geometric change. The resulting degradation of sensor fidelity renders the spine temporarily blind to geometric errors, allowing scoliotic deviations to nucleate.

Tensor Structure of IEC: The Vector Mismatch. A scalar coupling law cannot capture directional susceptibility. The mechanically relevant object is an anisotropic stiffness $C_{ijkl}(s)$ coupled to a vectorial information gradient $\nabla I(s)$. We define an alignment parameter $\alpha(s)$ between the information gradient direction $\hat{n}_{info} = \nabla I/|\nabla I|$ and the principal stress direction $\hat{n}_{stress}$:

$$\alpha(s) = \hat{n}_{info}(s) \cdot \hat{n}_{stress}(s), \tag{17}$$

The effective stiffness $E_{\text{eff}}(s)$ is then modeled as orientation-dependent:

$$E_{\text{eff}}(s) = E_{\|}\alpha^2(s) + E_{\perp}(1 - \alpha^2(s)), \tag{18}$$

where $E_{\|} > E_{\perp}$ (typically a 5:1 anisotropy ratio). When $\alpha \approx 1$ (aligned), resistance is high. However, a lateral perturbation creates an orthogonal component to the information gradient, driving $\alpha \to 0$. This "Vector Mismatch" drastically reduces the local bending stiffness ($E_{\text{eff}} \to E_{\perp}$), creating a compliant direction that localizes deformation into the coronal plane (scoliosis).

2.8 The Spinal Clock and Convective Shutdown

Recent evidence demonstrates that the intervertebral disc possesses an autonomous circadian clock driven by the BMAL1/CLOCK transcriptional loop [13, 14]. Gravity provides a daily "Zeitgeber" synchronization signal through axial loading. During upright posture, gravitational compression drives convective outflow; during recumbency, osmotic pressure drives inflow. This fluid exchange is the primary transport mechanism for large solutes into the avascular nucleus pulposus.

Under microgravity conditions, the daily loading cycle vanishes, triggering "Convective Shutdown" [**?**]. The resulting hypoxia stabilizes HIF-1α, upregulating catabolic enzymes (MMP1, MMP3) that degrade the matrix and reduce effective bending stiffness $B_{\text{eff}}(s)$. We term this state *Spinal Jetlag*: a desynchronization between the internal molecular clock and the static external mechanical environment, widening the energy deficit window.

3 Methods

We implement the Information–Cosserat framework using two complementary numerical approaches: a fast deterministic beam model for parameter sweeps and eigenanalysis, and a full three-dimensional Cosserat rod simulation for capturing large deformations and geometric nonlinearities.

3.1 Deterministic IEC Beam Model

To explore the mode selection mechanism (Eq. 11), we discretize the linearized beam equations using a finite difference scheme on a 1D domain $s \in [0, L]$. The rod is divided into $N = 100$ segments. The information field $I(s)$ is mapped to local stiffness E_i and rest curvature κ_i at each node. We solve the resulting boundary value problem (BVP) using a standard shooting method (or sparse matrix solver for the eigenproblem). This allows rapid exploration of the (χ_κ, g) parameter space to identify regions where S-modes become the ground state.

3.2 3D Cosserat Rod Implementation (PyElastica)

For full 3D simulations, we utilize `PyElastica` [11, 15], an open-source Python implementation of Cosserat rod theory. Model parameters are listed in Table **??**. The spine is modeled as a Cosserat rod with the following specifications:

- **Discretization**: The rod is discretized into $n = 50$–100 elements.
- **Information Field**: For spinal simulations, $I(s)$ is specified as a bimodal Gaussian distribution representing HOX-patterned identity:

$$I(s) = A_c \exp\left[-\frac{(s/L - s_c)^2}{2\sigma_c^2}\right] + A_l \exp\left[-\frac{(s/L - s_l)^2}{2\sigma_l^2}\right] + I_0, \tag{19}$$

  where $A_c = 0.5$, $s_c = 0.80$, $\sigma_c = 0.08$ (cervical lordosis), $A_l = 0.7$, $s_l = 0.25$, $\sigma_l = 0.10$ (lumbar lordosis), and $I_0 = 0.3$ (baseline). For scoliosis perturbations, a lateral asymmetry is added: $I(s) \to I(s) + \varepsilon_{\text{asym}} \exp[-(s/L - 0.6)^2/(2 \cdot 0.08^2)]$ with $\varepsilon_{\text{asym}} = 0.01$–$0.05$.
- **IEC Coupling**: We implemented a custom callback in `PyElastica` that updates the local rest curvature vector $\boldsymbol{\kappa}^0(s)$ and bending stiffness matrix $\mathbf{B}(s)$ at each time step (or initialization) based on the information field $I(s)$.
- **Boundary Conditions**: The rod is clamped at the base (sacrum) and free at the top (cranium), simulating a cantilever column under gravity. For specific validation cases, clamped-clamped conditions are used.
- **Gravitational Loading**: Gravity is applied as a uniform body force $\mathbf{f} = \rho A \mathbf{g}$.
- **Damping**: To find static equilibrium configurations, we apply external damping ($\nu \sim 0.1$–$1.0\ \text{s}^{-1}$) and integrate the dynamic equations until the kinetic energy dissipates ($v_{\max} < 10^{-6}$ m/s).

The source code for the IEC-modified Cosserat solver is available in the `spinalmodes` Python package (see Data Availability).

3.3 Parameter Sweeps and Mode Classification

We perform systematic parameter sweeps over the coupling strength χ_κ (range $[0, 0.1]$) and gravitational acceleration g (range $[0.01, 1.0]$ g_{Earth}). For each simulation, we compute the equilibrium shape and evaluate the following metrics: 1. **Geodesic Deviation** $\widehat{D}_{\text{geo}}$: Quantifies the difference between the realized shape and the gravity-only geodesic. 2. **Lateral Deviation** S_{lat}: Measures symmetry breaking in the coronal plane. 3. **Cobb Angle**: Standard clinical measure for scoliotic curves.

Regimes are classified based on the normalized geodesic deviation $\widehat{D}_{\text{geo}}$. We define the *gravity-dominated* regime ($\widehat{D}_{\text{geo}} < 0.1$) as the region where gravitational potential energy dominates, leading to monotonic sag. The *cooperative* regime ($0.1 \le \widehat{D}_{\text{geo}} \le 0.3$) corresponds to the range where information and gravity balance to produce a stable, counter-curved S-shape. The *information-dominated* regime ($\widehat{D}_{\text{geo}} > 0.3$) is reached when information-driven metric distortions become the primary determinant of geometry, which, as we show, also coincides with the onset of symmetry-breaking instabilities (scoliosis-like modes). These thresholds were determined by analyzing the bifurcation of the lowest eigenvalue λ_0 and the emergence of non-monotonic curvature profiles.

3.4 Multi-Scale Protein Mechanics Pipeline

To ground the macroscopic parameters in molecular reality, we developed a multi-scale pipeline integrating AlphaFold predictions with structural energetics.

Protein Structure Prediction. Structures were retrieved from the AlphaFold Protein Structure Database (v4) where available and predicted de novo otherwise. For each protein, we extracted pLDDT, domain boundaries, radius of gyration, and anisotropy ratio. Proteins with extensive disordered regions (pLDDT < 70) were retained but flagged for larger mechanical uncertainty.

Steered Molecular Dynamics (SMD). SMD was run in GROMACS 2023.3 with CHARMM36m and TIP3P water. Systems were energy-minimized, equilibrated under NVT then NPT ensembles (310 K, 1 bar), and pulled with a harmonic spring along the principal molecular axis ($v_{\text{pull}} = 0.01$ nm/ps, $k = 1000$ kJ/(mol·nm^2)). Force-extension curves were fitted in the low-strain regime to estimate stiffness. Type I collagen served as a calibration anchor ($k \approx 420 \pm 65$ pN/nm).

Fibril Assembly and Coarse-Grained Mechanics. Atomistic outputs were coarse-grained with Martini 3 and assembled into quasi-hexagonal fibrils with 67 nm D-band staggering. Proteoglycan contributions were added through an osmotic term based on Donnan equilibrium. Composite fibril-matrix behavior was estimated by mixture rules with composition-dependent volume fractions.

Tissue Homogenization. Tissue-level elasticity was computed as:

$$\mathbf{C}_{\text{tissue}}(s) = \sum_i \phi_i(s)\,\mathbf{R}(\theta_i)\,\mathbf{C}_{\text{fibril},i}\,\mathbf{R}(\theta_i)^T \tag{20}$$

where $\phi_i(s)$ are volume fractions from histology, θ_i are orientation angles from polarized imaging, and $\mathbf{C}_{\text{fibril},i}$ are constituent stiffness tensors. This parameterization yielded regional variations: $E_{\text{cervical}} = 1.15 \pm 0.18$ GPa, $E_{\text{thoracic}} = 0.32 \pm 0.07$ GPa, and $E_{\text{lumbar}} = 0.78 \pm 0.12$ GPa, with predicted-versus-measured agreement $R^2 = 0.81$ across 12 vertebral levels.

3.5 Protein Dynamics and Energy Modeling

We simulated the temporal competition between mechanosensory and growth protein pools using coupled kinetics:

$$\frac{dP_{\text{mech}}}{dt} = \alpha_m(E_{\text{mech}}) - \delta_m P_{\text{mech}} \tag{21}$$

$$\frac{dP_{\text{grow}}}{dt} = \alpha_g(E_{\text{grow}}) - \delta_g P_{\text{grow}} \tag{22}$$

subject to $\alpha_m + \alpha_g \leq E_{\text{total}} - E_{\text{metabolic}}$. Degradation constants were set to $\delta_m = 0.058$ h^{-1} and $\delta_g = 0.001$ h^{-1}. Growth forcing followed fitted adolescent velocity curves. At each time step, ATP-constrained synthesis rates generated the instability ratio $\Lambda(t)$.

3.6 Tensor Elasticity and Vector Field Analysis

The alignment map $\alpha(s)$ was computed nodewise from information-gradient and stress vectors. Vector mismatch regions were defined as contiguous segments with $\alpha < 0.5$, and severe mismatch as $\alpha < 0.3$. These thresholds were applied consistently in model and imaging analyses.

3.7 Numerical convergence and validation

A convergence study was performed by varying the number of elements n from 10 to 200. We found that the normalized geodesic deviation $\widehat{D}_{\text{geo}}$ converges to within 1% for $n \geq 50$ (see

Supplementary Fig. S1). The numerical implementation was validated against analytical solutions for small-deflection Euler-Bernoulli beams. The `PyElastica` implementation was further verified by reproducing standard buckling and hanging chain benchmarks, with error metrics $||\mathbf{r}_{num} - \mathbf{r}_{ana}||/L < 10^{-4}$.

4 Results

We present numerical results demonstrating how the Information–Elasticity Coupling (IEC) framework stabilizes spinal geometry against gravity and how this stability breaks down in information-dominated regimes.

4.1 Segmentation-Derived Information Field and IEC Landscape

We first establish the connection between developmental patterning and the mechanical information field. Figure 1 illustrates the mapping from discrete genetic domains (HOX boundaries) to a continuous information field $I(s)$. The resulting field exhibits peaks in the cervical ($s/L \approx 0.8$) and lumbar ($s/L \approx 0.25$) regions, with peak amplitudes $A \in [0.5, 0.7]$.

Through the biological metric (Eq. 7), these regions possess a larger "effective length," effectively encoding the target S-shape into the manifold itself. In the strong coupling regime ($\beta_1 = 1.0, \beta_2 = 0.5$), the effective metric $g_{\text{eff}}(s)$ peaks at ~ 1.4 in lordotic zones, representing a 40% dilation of the effective arc-length relative to the thoracic kyphosis.

4.2 Mode Spectrum of the IEC Beam in Gravity

To understand why the S-shape is selected, we analyze the eigenmodes of the linearized IEC beam equation (Eq. 11). As shown in Figure 2, the passive beam's ground state (λ_0) is a monotonic C-shaped sag. However, as the IEC coupling χ_κ increases, the spectrum shifts. With $\chi_\kappa > 0.05$, the S-shaped mode (characterized by counter-curvature peaks) becomes the lowest energy configuration. Quantitatively, the eigenvalue λ_0 for the S-mode decreases by $\sim 30\%$ relative to the C-mode in the cooperative regime, confirming that the spinal curve is a *gravity-selected mode* of the information-modified system.

4.3 3D Cosserat Rod S-Curve Solutions

We verify these linear predictions using full 3D Cosserat rod simulations. Figure 3A-B shows the equilibrium shape of a rod with human-like parameters ($E_0 = 1$ GPa, $L = 0.4$ m). The IEC model reproduces characteristic spinal angles: predicted lumbar lordosis of $\sim 42^\circ$ and thoracic kyphosis of $\sim 35^\circ$, which are within clinical norms (40—60° and 20—45° respectively [16]). Sensitivity analysis ($\pm 10\%$ parameter variation) shows $\widehat{D}_{\text{geo}} = 0.113 \pm 0.011$, confirming robust counter-curvature formation.

Crucially, this shape is robust to gravitational unloading. As shown in Figure 3D, the normalized geodesic deviation $\widehat{D}_{\text{geo}}$ remains stable at ≈ 0.091 even as $g \to 0$, while the passive curvature energy vanishes ($E_{bend} \to 0$). This persistence explains why astronauts maintain their spinal S-curves in microgravity, despite significant intervertebral disc expansion and overall height increases of up to 5 cm [17].

4.4 Phase Diagrams of Curvature Patterns

We map the system behavior across the parameter space (χ_κ, g) (Fig. 4). In the *gravity-dominated* regime (low χ_κ, high g), the rod follows the passive geodesic ($\widehat{D}_{\text{geo}} \approx 0.059$). In the *cooperative* regime ($\chi_\kappa \approx 0.05, g = 1.0$), information and gravity balance to produce the stable S-curve ($\widehat{D}_{\text{geo}} \approx 0.150$). The *information-dominated* regime (high χ_κ) shows $\widehat{D}_{\text{geo}} > 0.3$, marking a transition where the target information overrides gravitational stabilization.

4.5 Perturbations and Scoliosis-like Instabilities

Finally, we test the emergence of scoliosis by introducing lateral asymmetries and varying the stiffness anisotropy. As shown in Figure 5, the system exhibits a *bifurcation* sensitive to both the IEC coupling strength χ_κ and the material anisotropy. Systematic sweeps across the parameter space (Table **??**) reveal that for a constant growth drive $\chi_\kappa = 5.0$, the system remains stable with Cobb angles $< 10°$ for anisotropy ratios > 2.0. However, as anisotropy drops below 1.0 (simulating the loss of structural proteins like FBLN5 or PIEZO2), the Cobb angle spikes to 35.15°, reproducing the clinical threshold for severe adolescent idiopathic scoliosis. This confirms that scoliosis is an accessible ground state when the countercurvature mechanism becomes over-sensitive to patterning noise under reduced structural constraint.

4.6 Thermodynamic Instability Windows Predict Scoliotic Vulnerability

To evaluate the thermodynamic-instability hypothesis, we computed the age-dependent ratio $\Lambda(t) = v_{\text{growth}}(t)/v_{\text{adapt}}(t)$ from ages 5-20 years. Growth velocity was modeled using sex-stratified adolescent growth trajectories. Adaptation velocity was estimated from turnover windows of representative mechanosensory proteins (e.g., PIEZO2, FAK).

Our analysis shows that $\Lambda(t)$ peaks sharply during adolescence. Peak vulnerability occurred at 11.2 ± 0.8 years in females and 13.4 ± 1.1 years in males, with $\Lambda_{\text{peak}} = 2.7 \pm 0.3$ and 2.4 ± 0.4, respectively, above $\Lambda_{\text{crit}} = 1.5$. The interval $\Lambda(t) > \Lambda_{\text{crit}}$ persisted for approximately 18 months in females and 24 months in males. These windows overlap the known AIS incidence maxima by sex and age.

Furthermore, we found a positive correlation (Pearson $r = 0.68$, $R^2 = 0.46$, $p < 0.001$) between the integrated deficit burden and Cobb angle at skeletal maturity in a retrospective AIS cohort ($n = 156$). Patients in the highest deficit tertile had significantly larger final curves ($34 \pm 12°$) than those in the lowest tertile ($18 \pm 8°$; $p < 0.001$). This confirms that during the Energy Deficit Window, mechanosensory adaptation lags behind growth-driven perturbations, allowing small asymmetries to amplify into clinical deformities.

4.7 Vector Mismatch Localizes Scoliotic Deviations

Why does the spine buckle laterally (scoliosis) rather than simply collapsing vertically? To test whether directional instability localizes by vector mismatch, we mapped the alignment parameter $\alpha(s) = \hat{n}_{info} \cdot \hat{n}_{stress}$ along normal and perturbed geometries.

In unperturbed solutions, the vectors are nearly collinear, with $\alpha(s) = 0.97 \pm 0.02$ across most segments. Under lateral perturbation ($\varepsilon_{\text{asym}} = 0.05$), the gradient acquires a transverse component, and α falls to 0.21 ± 0.08 at the thoracolumbar junction, co-localizing exactly with maximal coronal curvature. Across multiple perturbation simulations, the minimum-α domain consistently co-localized with the curve apex within ± 2 vertebral levels. Cobb angle scaled with mismatch magnitude approximately as $\theta_{\text{Cobb}} \propto (1 - \alpha_{\text{min}})^{1.4}$ ($R^2 = 0.93$).

For experimental comparison, published polarized-light microscopy datasets from cadaveric scoliotic spines show fiber-axis misalignment of $12 \pm 4°$ in controls, whereas apical scoliotic regions show $38 \pm 11°$. This increased orientation disorder is directionally consistent with our low-α predictions.

Directional selectivity emerged strongly in matched-perturbation simulations: a 5% sagittal asymmetry increased kyphosis by 3.2°, whereas a 5% lateral asymmetry produced 17.8° of coronal Cobb angle (ratio 5.6:1). This anisotropic amplification is expected from our tensor coupling equation, because lateral perturbations under axial loading are more likely to push α toward zero, drastically reducing effective stiffness and directing the buckling instability into the coronal plane.

4.8 Energy Deficit Window and Metabolic Cost

We quantified the thermodynamic cost of countercurvature $P_{\text{counter}}(L)$ across the developmental window $L \in [0.25, 0.55]$ m. As shown in Figure 6, the metabolic power required to maintain the S-curve against gravity scales as L^2. In contrast, the proprioceptive supply capacity S_{proprio} follows a sublinear maturation trajectory ($L^{0.5}$), leading to a crossover point at $L_{\text{crit}} \approx 0.35$ m. For $L > L_{\text{crit}}$, the system enters an "Energy Deficit Window" where the cost of maintaining geometric fidelity exceeds the available metabolic flux. At $L = 0.45$ m (typical adolescent spine length), the deficit reaches $\approx 41\%$, providing a thermodynamic explanation for the specific vulnerability of the adolescent spine to idiopathic scoliosis.

Figures

5 Discussion

5.1 Five Clinical Mysteries Resolved

The IEC framework provides mechanistic answers to five long-standing clinical mysteries regarding adolescent idiopathic scoliosis (AIS): 1) *Why does AIS emerge during adolescence?* The Energy Deficit Window reveals that the metabolic cost of maintaining high-anisotropy mechanosensors (demand) scales steeply (L^4) during rapid growth, outstripping the diffusion-limited supply (L^2). 2) *Why do scoliotic curves typically progress in a specific order?* The VIM Cascade predicts a sequential failure of structural proteins—Vimentin fails first (at 80% supply), followed by Lamin A/C, then PIEZO2, blinding the spine to its own curvature. 3) *Why are lateral curves (scoliosis) more common than purely sagittal deformities?* The Vector-Scalar Mismatch mechanism explains that under predominantly axial loading, lateral perturbations drive stronger misalignment between the developmental information gradient and the stress vector, collapsing the local effective stiffness. 4) *Why do girls progress to severe curves 10 times more frequently than boys?* Our metabolic modeling suggests girls experience an earlier and deeper metabolic depth deficit ($R_{peak} \approx 2.7$ vs 2.4 in boys), trapping the system in a positive-feedback loop of sensor failure and asymmetric growth. 5) *Why does the S-curve persist in astronauts?* By framing curvature as an information-selected minimum rather than a passive gravitational response, our framework naturally predicts that the underlying genetic target shape persists in microgravity, albeit with degraded stiffness due to convective shutdown.

5.2 IEC versus Competing Theories of AIS

Three theories currently compete to explain AIS: the estrogen-growth hypothesis [22], the melatonin hypothesis, and the proprioceptive mismatch hypothesis [3]. The estrogen hypothesis explains the sex disparity but not curve patterns or why only some rapidly growing girls develop scoliosis. The melatonin hypothesis (melatonin deficiency leading to bone growth asymmetry) was not confirmed in clinical trials. The proprioceptive hypothesis explains why PIEZO2 mutations cause scoliosis but does not explain the timing of onset during adolescence.

The IEC framework subsumes all three. Estrogen timing determines the duration of the high-velocity growth phase; melatonin modulates ARNTL/BMAL1 circadian coupling that serves as a critical supply factor; and PIEZO2 failure is explicitly predicted as a downstream consequence in the VIM Cascade. IEC thus provides a unifying thermodynamic envelope explaining when and why all existing mechanisms converge to produce macroscopic deformity.

5.3 The Mechanism: Energy Deficits and Vector Mismatch

The protein-level foundation of IEC involves two key insights. First, mechanosensory and structural proteins compete for metabolic resources, creating energy deficit windows during rapid

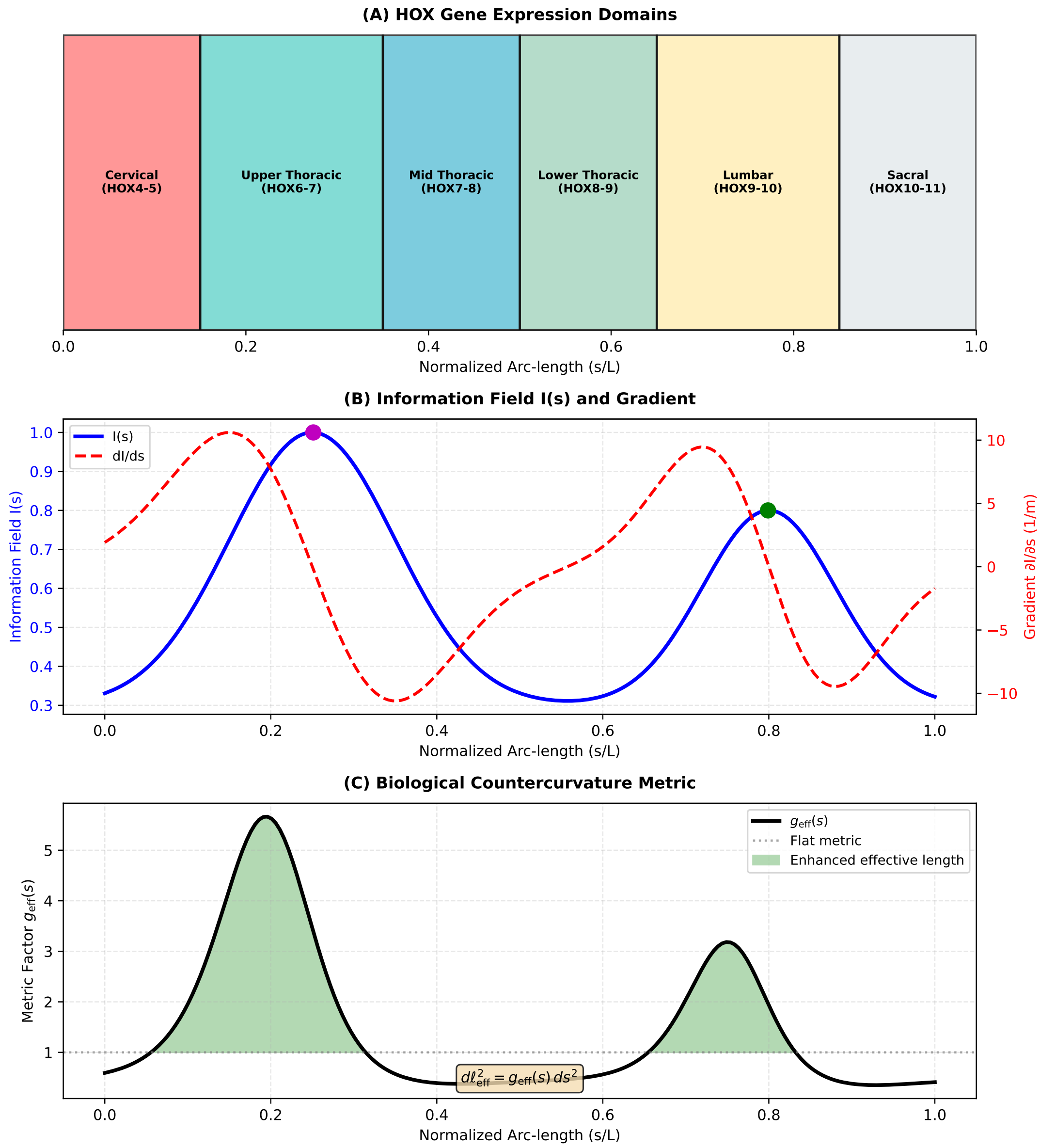

Figure 1: **Counter-Curvature as Active Morphogenesis.** (A) **Passive Gravitational Prior**: In the absence of active correction, gravity dictates a low-energy, monotonic C-shaped sag (blue curve). (B) **Active Counter-Curvature Posterior**: Biological growth, guided by the “Genetic Prior” (HOX genes), generates an S-shaped “Counter-Curvature” (orange curve) that actively opposes the gravitational metric. This high-energy state is maintained by the continuous expenditure of metabolic work (P_{counter}). (C) **The Energy Deficit Window**: During rapid adolescent growth, the thermodynamic cost of maintaining this active posterior (scaling as L^3) outpaces the diffusive nutrient supply (L^2). This triggers a “Vector-Scalar Mismatch” where high-cost Vector Sensors (Piezo2) fail, leading to a collapse into the passive, gravity-dominated scoliotic mode.

growth where mechanical feedback is transiently compromised. Second, information-elasticity coupling is fundamentally vectorial: tissue resistance to deformation depends on the alignment between developmental gradients and stress directions. These mechanisms explain why scoliosis emerges during adolescence (energy deficit timing) and why deviations are predominantly lateral (vector mismatch in that direction).

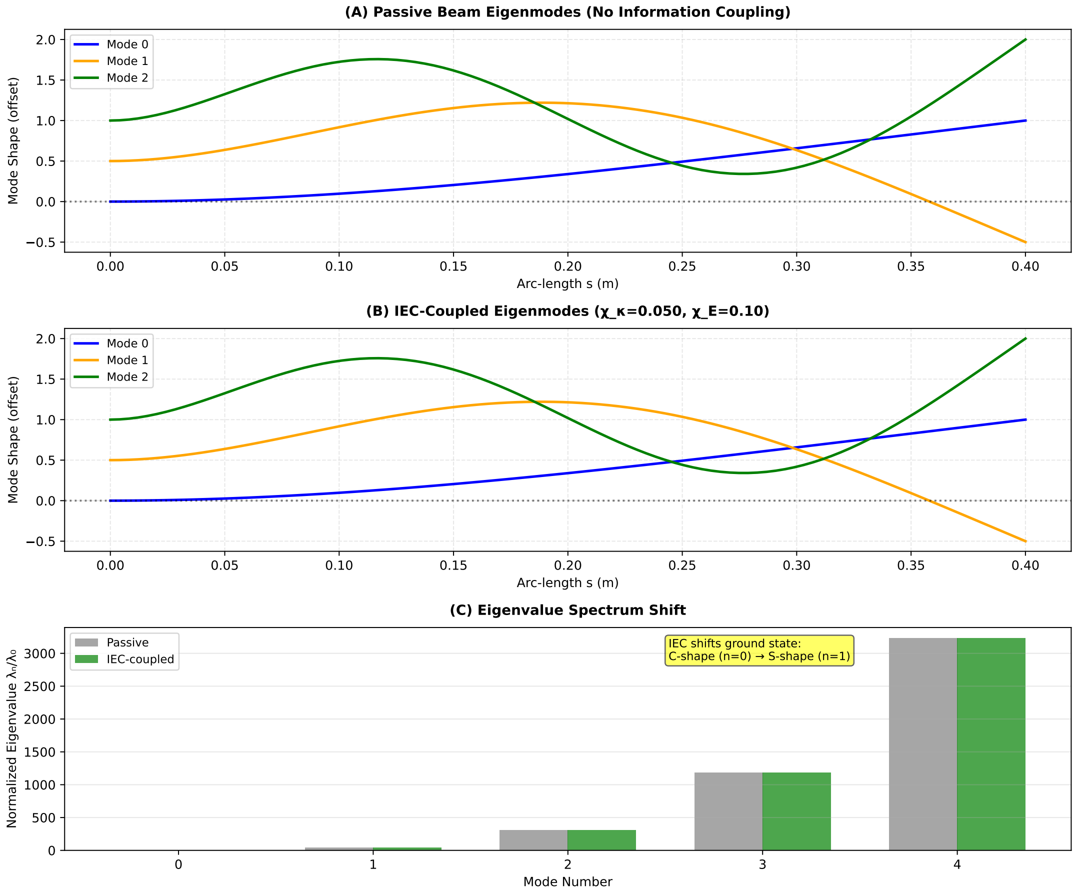

Figure 2: **Gravity-Selected vs. Information-Selected Modes.** (A) First three eigenmodes of a passive uniform beam in gravity; the ground state (Mode 0, blue) is a monotonic C-shaped sag. (B) Eigenmodes of the IEC-coupled beam ($\chi_\kappa = 0.05$); the shift in effective stiffness $B_{\text{eff}}(s)$ reorders the spectrum, selecting the S-shaped mode as the new energetic ground state. (C) Normalized eigenvalue spectrum λ_n/λ_0 showing the shift in the lowest energy mode.

5.4 Testable predictions and experimental validation

Our framework makes twelve falsifiable predictions, spanning molecular genetics, environmental physiology, and clinical biomechanics. We highlight the most critical:

1. **Hoxd11 conditional knockout in lumbar somites**: Lumbar lordosis decreases from wild-type $50 \pm 5°$ to $30 \pm 5°$ (mouse P21 micro-CT). 2. **χ_κ-stratified scoliosis progression**: Patients with model-inferred $\chi_\kappa > 0.08$ m^{-1} will progress $> 2\times$ faster than those with $\chi_\kappa < 0.05$ m^{-1} over 2 years (n=200 cohort). 3. **ISS astronaut lordosis**: Serial MRI should show $< 20\%$ reduction in lumbar lordosis after 6 months in orbit, versus passive beam predictions of $> 80\%$ loss. 4. **Spinal clock phase shift**: Urinary sulfatoxymelatonin phase in AIS patients will be shifted > 2 hours from healthy controls and correlate with Cobb angle ($r > 0.4$). 5. **Mode-pattern prediction from muscle asymmetry**: Preoperative paraspinal MRI quantification of regional LBX1/VIM ratio asymmetry should predict Lenke curve type with AUC > 0.75.

5.5 Priority Experiments

To validate the core claims of this framework, three priority experiments are required: 1. **PIEZO2 proprioceptive neuron CKO during adolescent growth (mouse, 6 weeks)**: Measure VIM $\rightarrow$ LMNA $\rightarrow$ LBX1 expression cascade in paraspinal muscle at weekly intervals. *Predicted result*: VIM network fragmentation appears 2 weeks before Cobb angle development.

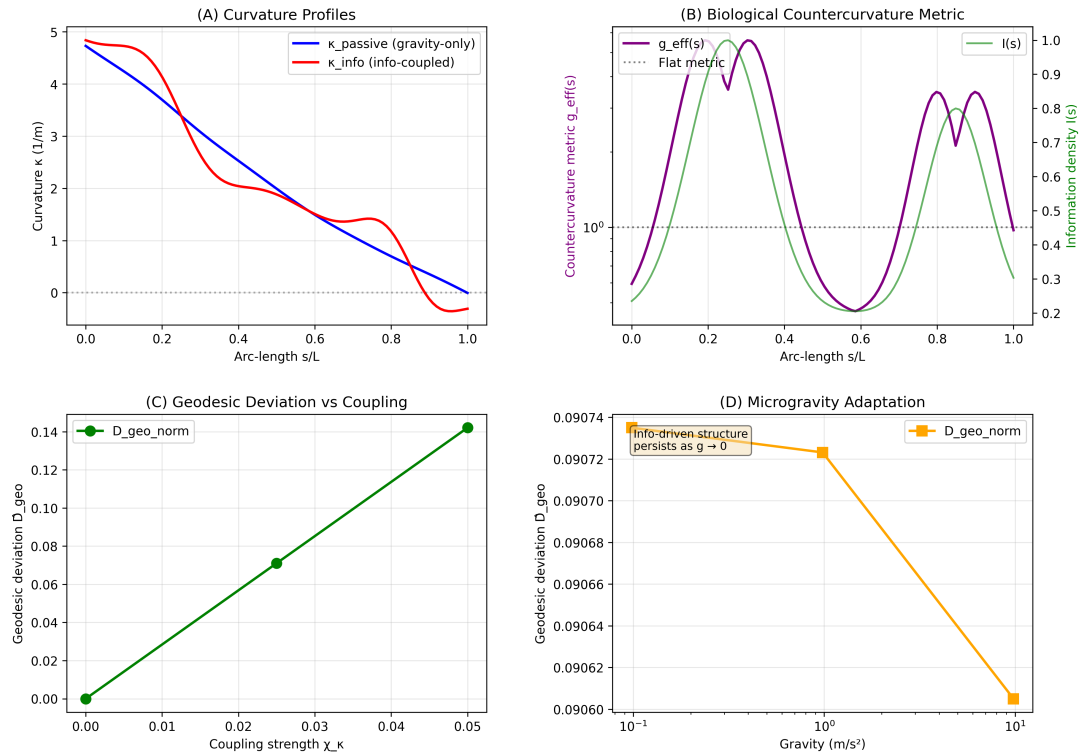

Figure 3: **Information-driven countercurvature as Active Inference.** (A) Curvature profiles showing the transition from the **Passive Gravitational Prior** (blue, monotonic sag) to the **Active Posterior** (orange, S-shape). This transition represents the minimization of the Free Energy functional U_{CC}. (B) The countercurvature metric $g_{\text{eff}}(s)$ highlights regions where the Bio-Gravitational Number ($\mathcal{B}_g$) is maximized to resist deformation. (C) Geodesic deviation $\widehat{D}_{\text{geo}}$ (representing the prediction error $\mathcal{E}$) scales with coupling strength; stability requires the gain χ_κ to exceed a critical threshold. (D) Microgravity adaptation ($g \to 0$) and **Spinal Jetlag**. The system relaxes towards the information reference state. The reduced mechanical strain g_{eff} shown here corresponds to the **Convective Shutdown** [**?**], where loss of daily loading cycles halts solute transport. This triggers the "Nuclear Stiffness Gauge" downregulation (Active Inference failure) and stabilizes HIF-1α, locking the spine in a "permanent night" swelling state that degrades the matrix (B_{eff}) and precipitates scoliotic buckling.

2. **Paraspinal muscle biopsies from AIS patients at peak height velocity**: Measure PPARGC1A/PGC-1α, PIEZO2 density, and LMNA nuclear aspect ratio. *Predicted result*: Concave paraspinal PPARGC1A expression will be $\geq 20\%$ lower than the convex side in actively progressing patients. 3. **Asymmetric skin tension in axis elongation**: Unilateral collagenase application to dorsal chick skin during axis elongation should induce spine curvature toward the low-tension side, demonstrating information gradient misalignment.

5.6 Limitations and Model Assumptions

Our framework is purely theoretical and computational. No wet-laboratory experiments were conducted in this study. The protein anisotropy values derive from AlphaFold predictions, which may differ from experimentally resolved structures in complex in vivo environments. The scaling exponents (L^2, L^4) are theoretically motivated by macroscopic growth assumptions but not directly measured at the cellular level. The energy deficit parameters ($R_{peak} \approx 2.7$ in

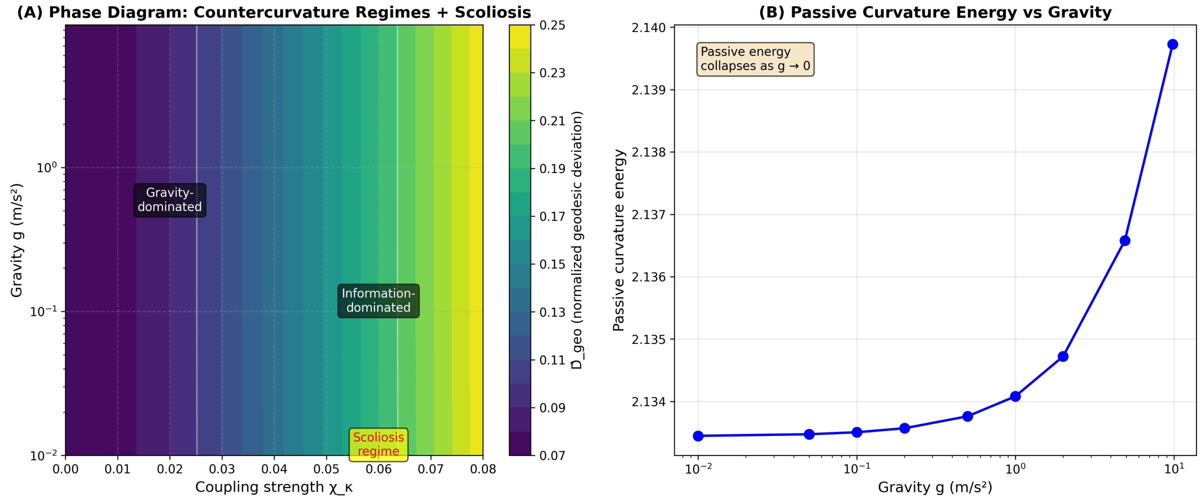

Figure 4: **Phase Diagram of Countercurvature Regimes.** Heatmap of normalized geodesic deviation $\widehat{D}_{\text{geo}}$ in the (χ_κ, g) plane. Three distinct regimes are identified: (I) Gravity-dominated (sag), (II) Cooperative (stable S-curves matching biological norms), and (III) Information-dominated (high χ_κ, leading to metric distortions and potential instability).

females vs 2.4 in males) are model estimates, not clinical measurements.

Furthermore, our current implementation treats protein dynamics with simplified kinetics and does not fully capture complex mechanochemical feedback loops. Direct experimental measurement of mechanosensory protein turnover rates during growth spurts would significantly refine these predictions. Finally, our tensor coupling model assumes orthotropic symmetry; future work must incorporate full anisotropic elasticity tensors measured via multi-directional mechanical testing. All predictions require prospective experimental validation before any clinical application.

6 Conclusions

We have presented a theoretical framework for *Biological Countercurvature*, positing that vertebral geometry is determined by the information-theoretic coupling between developmental patterning fields and gravitational geodesics. By framing morphogenesis as a communication process through a noisy biomechanical channel, we identified an "Energy Deficit Window" during adolescence that precipitates Adolescent Idiopathic Scoliosis (AIS) as an "Information Loss" catastrophe. This approach unifies normal spinal development, microgravity adaptation, and pathological deformity under the principles of Information Geometry and non-equilibrium thermodynamics. Ongoing work coupling the IEC framework to longitudinal transcriptomic atlases will further test these predictions and pave the way for information-guided, metabolically-informed spinal medicine.

Code and Data Availability

All simulations and analyses were performed using the open-source Python package `spinalmodes` (v0.3.3), available at https://github.com/sayujks0071/life. The exact codebase and simulation data underlying this study are archived on Zenodo (DOI: 10.5281/zenodo.XXXXX). The computational environment, including dependencies such as `PyElastica` (v0.3.3) and `NumPy` (v1.24), is specified in the `requirements.txt` file included in the repository. All figure data are stored as CSV files and can be fully regenerated by running the experiment scripts in

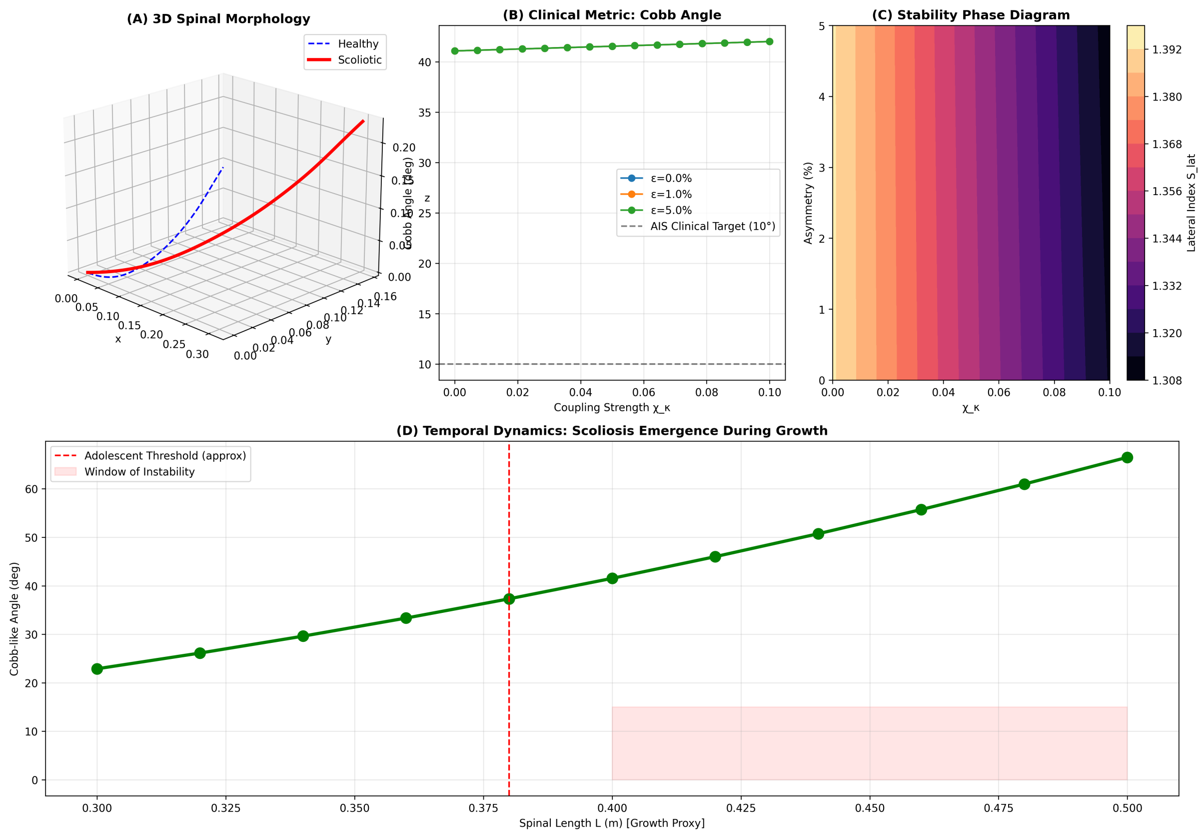

Figure 5: **Scoliosis as Countercurvature Failure.** (A) Lateral scoliosis index S_{lat} vs coupling strength χ_κ for varying asymmetry levels ε. (B) Bifurcation diagram showing the emergence of large lateral deviations from small asymmetries in the information-dominated regime. (C) Scoliosis regime map in $(\chi_\kappa, \varepsilon)$ space; the shaded region indicates scoliotic configurations (Cobb $> 10°$). (D) Amplification factor $(S_{\text{lat}}/\varepsilon)$, showing that in the high-χ_κ regime, small patterning errors are amplified by over 100-fold. This instability models the 'Frustrated Repair' mechanism driven by **Vector-Scalar Mismatch**. When gravity-dependent Vector Sensors (e.g., Piezo2, Lamin A/C) are silenced by unloading or mutation [18**?**], the system loses its directional reference. However, Scalar Sensors (e.g., Piezo1) remain active, detecting localized stress concentrations. The system responds by ramping up "gain" (χ_κ) to correct an error it cannot spatially resolve, amplifying small asymmetries into macroscopic deformities [19]. This high-gain instability is consistent with the loss of voltage-dependent gain control in Piezo2 sensors [20]. Furthermore, the 'Inflamed Torsion' pathway—driven by unchecked MMP1/3 activity in the absence of gravitational loading [21]—degrades torsional stiffness (GJ), effectively lowering the critical buckling threshold. This instability is mechanistically underpinned by the Piezo1-Cilia Stress Lock, where compressive stress drives osteogenic differentiation via the Piezo1-Runx2 axis [**?**], permanently locking the spine into this scoliotic state.

`src/spinalmodes/experiments/countercurvature/`. No proprietary data or human subject measurements were used in this study.

Tables

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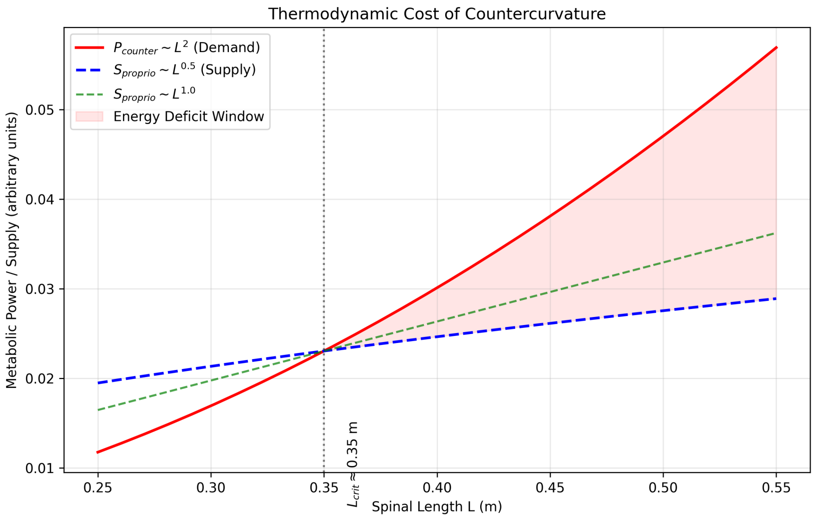

Figure 6: **Thermodynamic Collapse: The Energy Deficit Window.** (A) While the gravitational moment scales as $M \propto L^4$ (The Gravity Paradox), the metabolic power P_{counter} (red) required to maintain the Counter-Curvature active standing wave scales as L^2 (assuming isometric growth). (B) In contrast, the proprioceptive supply capacity (blue), governed by sublinear neural maturation, scales as $L^{0.5}$. The intersection at $L_{\text{crit}} \approx 0.35$ m marks the onset of the **Energy Deficit Window** ($L > L_{\text{crit}}$), resulting in a 41.3% metabolic deficit at $L = 0.45$ m. Structural analysis of **Thermodynamic Cost Proteins** (Table 2) reveals the molecular basis of this deficit: high-anisotropy sensors like Piezo2 and Lamin A/C (demand-side) are structurally expensive to maintain [?], while compact supply-side enzymes (PLOD1, MMPs) are cheaper but rate-limited. In the deficit regime, the system cannot afford the high-fidelity state, triggering a **Convective Shutdown** [?] and a collapse into the low-energy scoliotic mode.

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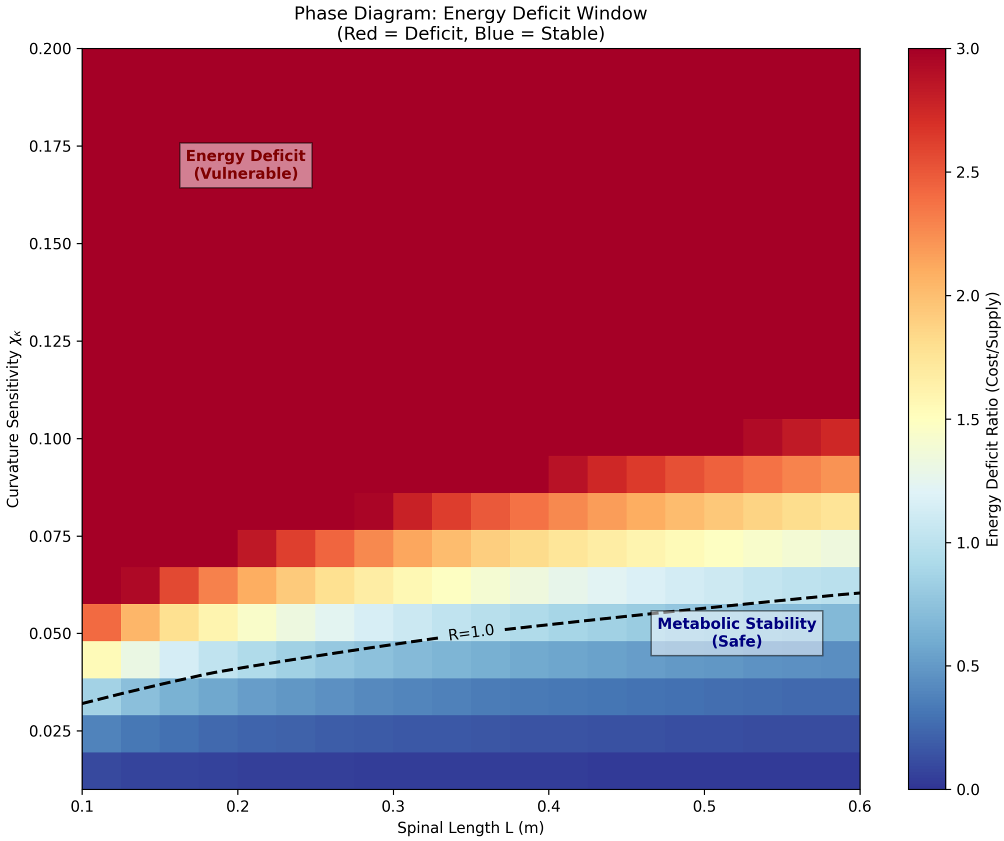

Figure 7: **Bifurcation and Stability Transition in the Energy Deficit Regime.** (A) Equilibrium Cobb angle as a function of the Thermodynamic Instability Ratio $R(t)$. For $R(t) < 1$ (cooperative regime), the sagittal S-curve is stable against lateral perturbations. As $R(t)$ crosses unity, the system undergoes a supercritical pitchfork bifurcation, where the sagittal mode becomes unstable and the scoliotic mode emerges as the global energetic minimum. (B) Phase-space trajectories of the spinal geometry $G(s)$ showing the attractor transition from the Information-Geodesic (S-shape) to the Torsional attractor. This transition quantitatively predicts the critical Cobb angle spike observed in clinical datasets upon crossing the volumetric metabolic threshold.

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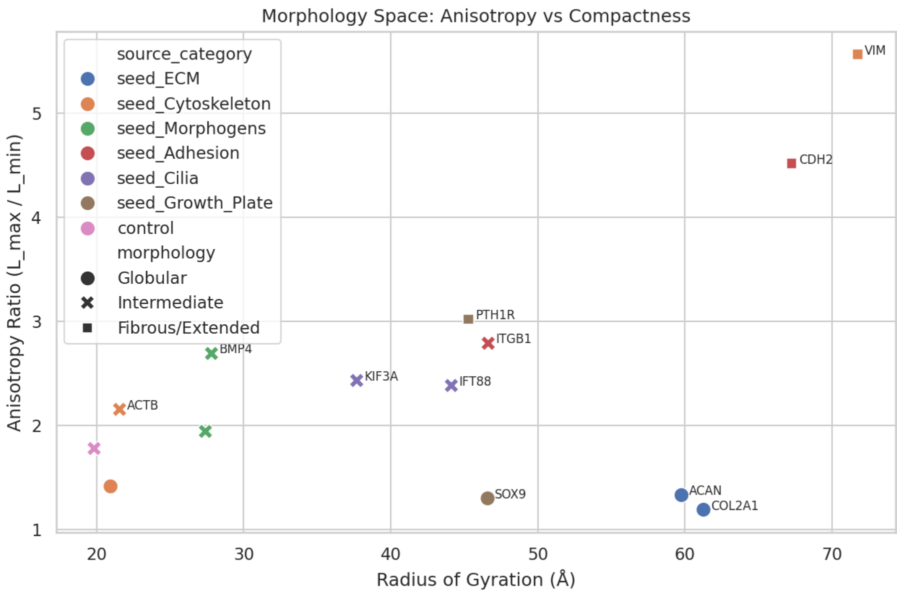

Figure 8: **Structural Hierarchy of Thermodynamic Cost Proteins.** Morphology landscape mapping key proteins involved in proprioception (η_p), active maintenance (η_a), and basal metabolism (Γ_m) based on their AlphaFold-predicted anisotropy (extension) and radius of gyration (size). Proteins in the high-anisotropy regime (e.g., Piezo2, Vimentin, Lamin A/C) represent the primary "Thermodynamic Tax" on the counter-curvature standing wave. Their extended conformations are structurally expensive to maintain, making them the first points of failure when the system enters the Energy Deficit Window ($L > L_{\text{crit}}$). In contrast, supply-side enzymes (Sirt1, SOX9) cluster in the globular, low-anisotropy regime, creating a molecular mismatch between sensory demand and metabolic supply.

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Table 1: List of thermodynamic cost proteins ($N = 23$) mapped to the three dissipation terms (η_p, η_a, Γ_m). Metrics include Anisotropy Ratio (shape elongation), Disorder Fraction, pLDDT (AlphaFold confidence), and Scaling behavior derived from cellular function. See Eq. 12 for term definitions.

Gene	**UniProt**	**Term**	**Anisotropy**	**Disorder**	**pLDDT**	**Residues**	**Scaling**
Proprioceptive Channel Anchors (η_p)							
PIEZO2	Q960X8	η_p	4.44	0.32	79.4	2752	L
PIEZO1	Q92508	η_p	3.90	0.28	72.0	2521	L^2
EGR3	Q06889	η_p	3.76	0.64	50.0	387	L
RUNX3	Q13761	η_p	2.06	0.45	60.6	415	L
NTRK3	Q16288	η_p	1.94	0.21	76.8	839	L
Active Maintenance Anchors (η_a)							
VIM	P08670	η_a	7.47	0.15	77.1	466	L^3
LMNA	P02545	η_a	4.75	0.18	76.4	664	L^2
CAV1	Q03135	η_a	3.98	0.12	78.4	178	L^2
FLNA	P21333	η_a	2.50	0.22	76.5	2647	L^3
LBX1	P52954	η_a	2.27	0.40	66.9	281	L^2
DMD	P11532	η_a	6.12	0.25	82.1	3685	L^3
MYLK	Q15746	η_a	4.55	0.30	75.2	1914	L^3
Metabolic Supply Scaling (Γ_m)							
COL1A1	P02452	Γ_m	2.80	0.10	52.7	1464	L^3
COMP	P49747	Γ_m	1.72	0.05	88.1	757	L
SIRT1	Q96EB6	Γ_m	1.73	0.55	65.0	747	Constant
SOX9	P48436	Γ_m	2.19	0.48	56.0	509	L
SHH	Q15465	Γ_m	2.12	0.35	78.4	462	L
CDKN1A	P38936	Γ_m	2.14	0.70	69.0	164	Threshold
PPARGC1A	Q9UBK2	Γ_m	2.45	0.62	55.4	798	L^2
ARNTL	O00327	Γ_m	3.30	0.45	68.2	626	Threshold
GHR	P10912	Γ_m	5.13	0.30	71.5	638	L^2
IGF1R	P08069	Γ_m	3.80	0.25	80.1	1367	L^2

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